frequency domain, and a negative sine wave being added to the imaginary part. The *amplitude* of each sinusoid is given by the *amplitude* of the time domain sample. The *frequency* of each sinusoid is provided by the *sample number* of the time domain point. The algorithm involves: (1) stepping through each time domain sample, (2) calculating the sine and cosine waves that correspond to each sample, and (3) adding up all of the contributing sinusoids. The resulting program is nearly identical to the correlation method (Table 8-2), except that the outer and inner loops are exchanged.

The Sinc Function

Figure 11-4 illustrates a common transform pair: the *rectangular pulse* and the *sinc function* (pronounced "sink"). The sinc function is defined as: $\text{sinc}(a) = \sin(\pi a)/(\pi a)$, however, it is common to see the vague statement: "the sinc function is of the general form: $\sin(x)/x$." In other words, the sinc is a sine wave that decays in amplitude as $1/x$. In (a), the rectangular pulse is symmetrically centered on sample zero, making one-half of the pulse on the right of the graph and the other one-half on the left. This appears to the DFT as a single pulse because of the time domain periodicity. The DFT of this signal is shown in (b) and (c), with the *unwrapped* version in (d) and (e).

First look at the unwrapped spectrum, (d) and (e). The *unwrapped magnitude* is an oscillation that decreases in amplitude with increasing frequency. The phase is composed of all zeros, as you should expect for a time domain signal that is symmetrical around sample number zero. We are using the term *unwrapped magnitude* to indicate that it can have both positive and negative values. By definition, the *magnitude* must always be positive. This is shown in (b) and (c) where the magnitude is made all positive by introducing a phase shift of π at all frequencies where the unwrapped magnitude is negative in (d).

In (f), the signal is shifted so that it appears as one contiguous pulse, but is no longer centered on sample number zero. While this doesn't change the magnitude of the frequency domain, it does add a linear component to the phase, making it a jumbled mess. What does the frequency spectrum look like as real and imaginary parts ? Too confusing to even worry about.

An N point time domain signal that contains a unity amplitude rectangular pulse M points wide, has a DFT frequency spectrum given by:

EQUATION 11-1
DFT spectrum of a rectangular pulse. In this equation, N is the number of points in the time domain signal, all of which have a value of zero, except M adjacent points that have a value of one. The frequency spectrum is contained in $X[k]$, where k runs from 0 to $N/2$. To avoid the division by zero, use $X[0] = M$. The *sine* function uses radians, not degrees. This equation takes into account that the signal is *aliased*.

$$Mag\ X[k] = \left| \frac{\sin(\pi k M/N)}{\sin(\pi k/N)} \right|$$